dig cheat sheet

Everyday commands

Command	What it does
`dig example.com`	A record, full output
`dig example.com MX`	Another type: AAAA, NS, MX, TXT, SOA, CAA…
`dig example.com +short`	Values only
`dig example.com +noall +answer`	Answer section only, with TTLs: the best short form
`dig @1.1.1.1 example.com`	Ask this server, not your default resolver
`dig example.com +trace`	Walk the tree from the root, like a resolver
`dig -x 8.8.8.8`	Reverse lookup (PTR)
`dig SOA example.com +multiline`	SOA fields, one per line, with comments
`dig -f list.txt TXT +noall +answer`	One lookup per line of the file
`dig example.com +tcp`	Use TCP instead of UDP
`dig example.com +dnssec`	Also show the signatures (RRSIG)
`dig @127.0.0.1 -p 5300 example.com`	Query a non-standard port

Ask the authoritative server directly (raw TTL, `aa` flag):
`dig NS example.com +short`, then
`dig @<one of the NS> example.com +norecurse`

Get the answer for another location (EDNS Client Subnet, needs a resolver that supports it):
`dig @8.8.8.8 google.com +subnet=129.20.0.0/24`

Make the short form the default: put `+noall +answer` in `~/.digrc`. Remove it to get the full output back.

Without dig

Tool	Example
nslookup	`nslookup -type=MX example.com 1.1.1.1`
PowerShell	`Resolve-DnsName example.com -Type MX -Server 1.1.1.1`
Browser	`toolbox.googleapps.com/apps/dig`

Reading the output

```
❯ dig www.example.com
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 17148
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, …

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;www.example.com.          IN A

;; ANSWER SECTION:
www.example.com.       300 IN CNAME a1422.dscr.akamai.net.
a1422.dscr.akamai.net.  20 IN A     23.72.250.233

;; Query time: 185 msec
;; SERVER: 192.168.1.1#53(192.168.1.1) (UDP)
```

An answer line reads: **name · TTL · class · type · value**.

Section	Contains
HEADER	Status, flags, number of records per section
OPT	EDNS: UDP answers over 512 bytes, cookie, client subnet
QUESTION	What you asked: name, class, type
ANSWER	The records
AUTHORITY	The zone's NS (a referral) or SOA (a negative answer)
ADDITIONAL	Extra records, e.g. the IP of a listed NS (glue)
Footer	Server that answered, time, transport, size

Status codes

Status	Meaning
`NOERROR`	The query worked
`NXDOMAIN`	This name does not exist
NODATA	The name exists, but has no record of this type: `NOERROR` with `ANSWER: 0`
`SERVFAIL`	No valid answer, e.g. DNSSEC failure, servers unreachable
`REFUSED`	The server declines to answer (policy)

Flags

Flag	Meaning
`qr`	This is a response
`aa`	Authoritative answer: from the zone's own server
`rd`	Recursion desired (set by the client)
`ra`	Recursion available (the server is a resolver)
`tc`	Truncated: retry over TCP
`ad`	Validated by DNSSEC, according to the resolver

Record types

Type	Holds
`A` / `AAAA`	IPv4 / IPv6 address
`CNAME`	An alias: resolution restarts with the new name
`MX`	Mail servers, lowest preference number first
`NS`	Name servers of the zone
`SOA`	Zone origin: primary server, serial, timers, negative TTL
`TXT`	Free text: SPF, domain verification
`PTR`	Name of an IP, under `in-addr.arpa` or `ip6.arpa`
`SRV`	Host and port of a service, with priority
`CAA`	Certificate authorities allowed to issue for the domain
`DNSKEY` `DS` `RRSIG`	DNSSEC: keys, delegation, signatures

TTL and caches

- **From a resolver**, the TTL is the time left in its cache. **From the authoritative server** (`@ns +norecurse`), it is the configured value.
- **Negative answers are cached too**, for the lower of the SOA's TTL and its last field (MINIMUM).
- **Lookup order:** browser cache → OS cache and hosts file → recursive resolver → root, TLD, authoritative.
- **Before a change**, lower the TTL at least one old TTL in advance.

Flush the caches

Where	Command
Windows	`ipconfig /flushdns` (`/displaydns` lists it)
Linux	`resolvectl flush-caches`
macOS	`sudo dscacheutil -flushcache` `sudo killall -HUP mDNSResponder`
Chrome	`chrome://net-internals/#dns`
Firefox	`about:networking#dns`